

Advanced International Macroeconomics B (EI136)

Class 8

Dominant currencies

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April 14, 2026

Outline of today class

- Dominant currencies: patterns (ECB 2025).
- International trade invoicing and impact on macroeconomic transmission.
 - Exchange rate pass-through (Boz et al. 2025, 2022, Gagnon and Sarsenbayev 2023).
 - Invoicing of services (Amador et al. 2024).
 - Endogenous invoicing (Mukhin 2022).
- Financial market considerations.
 - Cost of foreign currency debt ('Casas, Meleshchuk, and Timmer 2022).
 - Heterogeneity in currency of bonds (Maggiori et al. 2020).
- Sensitivity to fragmentation (Chahrour et al 2024).

DOMINANT CURRENCIES : PATTERNS

Dominant role of the dollar

- The dollar represents a large share of international activity: exchange reserves, bonds, loans, FX turnover, (ECB 2025).
 - Sizable role of the euro in payments.
- Declining dollar role in central banks' reserves, with rising role of several secondary currencies (not a renminbi story).

Overview of the international roles

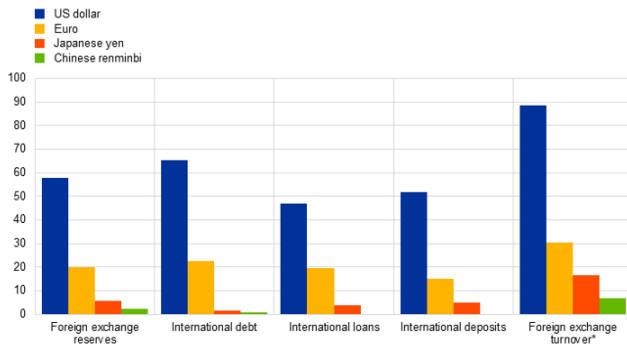
- Sizable dollar role in many dimensions.

Chart 2

The euro remained the second most important currency in the international monetary system

Snapshot of the international monetary system

(percentages)



European Central Bank (2025). "The International Role of the Euro".

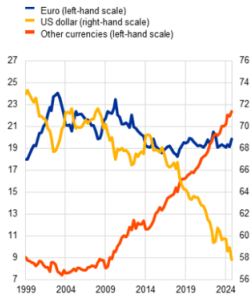
- Decreasing dollar share, with a move towards “non traditional” reserves (Australia, Korea, New Zealand).

Chart 4

Stable euro share in global foreign exchange reserves and declining US dollar share to the benefit of non-traditional reserve currencies

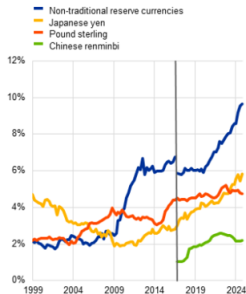
a) Share of the euro and US dollar in global foreign exchange reserves

(percentages; at constant Q4 2024 exchange rates)



b) Share of other currencies in global foreign exchange reserves

(percentages; at constant Q4 2024 exchange rates)



European Central Bank (2025). “The International Role of the Euro”.

Impact of a dominant currency

- Gains: lower transaction cost, lower cost of debt (exorbitant privilege), stronger international effects of policy with positive “spillbacks”, less exposure to foreign shocks on import prices (low exchange rate pass-through).
- Costs: safe haven pattern with appreciation in crises (exorbitant duty), global responsibility (provision of swap lines), less impact of policy on import prices (low exchange rate pass-through), capital flows volatility (limited).
- Evidence for lower cost of sovereign debt. Holdings by foreign central banks lead to a lower term premium (reserves are a source of demand that is not price sensitive).

INVOICING OF INTERNATIONAL TRADE

The dollar and international trade

- The dollar is largely used in invoicing of international trade, much more than the share of the US in exports and imports (Boz et al. 2025).
 - Broad data set with evidence on invoicing currencies, including Renminbi (132 countries over 1990-2023).
- Large dollar invoicing role, well above the role of the U.S. as a trade partner.
 - Invoicing role of the euro more in line with its trade role.
- The renminbi share is gradually increasing, but remains regionally concentrated.
- Panel analysis of drivers of (country-level) invoicing shares.
 - Share of trade with the issuing country.
 - Share of trade in commodities favors the dollar: “coalescing” effect.

- Large dollar share, gradual increase of renminbi.

With share of exports to China and renminbi invoicing share

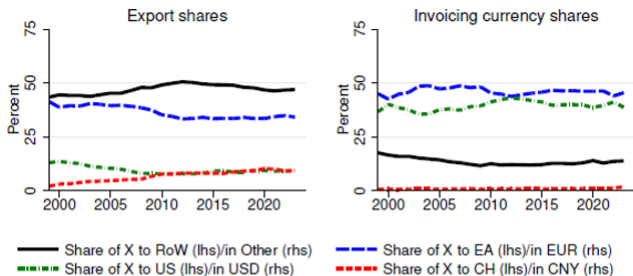
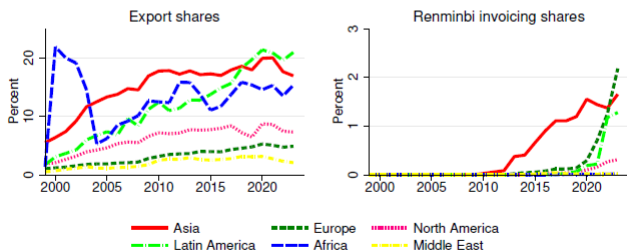


Figure 5: Zooming in on exports to China and renminbi invoicing



Boz, Emine, Anja Brügger, Camila Casas, Georgios Georgiadis, Gita Gopinath, and Arnaud Mehl (2025). "Patterns of Invoicing Currency in Global Trade in a Fragmenting World Economy", IMF working paper 25/178

Drivers of invoicing

- Trade with the issuer country, and in commodities.

Table 2: Panel regressions for compositional/mechanical correlates of export invoicing shares

	USD		EUR		CNY	
	(1)	(2)	(3)	(4)	(5)	(6)
Lagged invoicing currency share	0.73*** (0.00)	0.75*** (0.00)	0.80*** (0.00)	0.80*** (0.00)	0.92*** (0.00)	1.02*** (0.00)
Bilateral trade share with invoicing currency issuer	0.11*** (0.00)	0.06* (0.07)	0.08*** (0.00)	0.09*** (0.00)	0.01*** (0.00)	0.01*** (0.00)
Trade share with rest of invoicing currency block	0.02 (0.53)	0.06** (0.04)	0.03 (0.23)	0.03 (0.24)	0.01** (0.02)	0.01* (0.06)
Bilateral exchange rate against invoicing currency	-0.00 (0.87)	0.00 (0.97)	-0.01** (0.03)	-0.00 (0.29)	0.00** (0.04)	0.00* (0.08)
Commodity trade share	0.09*** (0.00)		-0.04** (0.04)		-0.00 (0.74)	
Oil trade share		0.17*** (0.01)		-0.16*** (0.00)		-0.01** (0.04)
Year FEs	✓	✓	✓	✓	✓	✓
Within R-squared	0.65	0.65	0.77	0.78	0.80	0.96
Observations	1298	1307	1322	1321	724	712
Countries	114	111	113	111	91	89

Boz, Emine, Anja Brügger, Camila Casas, Georgios Georgiadis, Gita Gopinath, and Arnaud Mehl (2025). "Patterns of Invoicing Currency in Global Trade in a Fragmenting World Economy", IMF working paper 25/178

- Geopolitics are now more central, with Russia-Ukraine war.
- Sanctions have shifted invoicing of Russia trade towards renminbi.
- Panel econometrics show that geopolitical distance matters (measured by proximity in UN General Assembly votes).
 - For the dollar and the euro, the effect is post 2021. Geopolitics have only recently become an issue.
- “Rotation” between Dollar / Euro and reniminbi.

Russia shift

- Russia trade has moved towards China, with renminbi invoicing.

Chart 13

Increasing use of the renminbi in China's and Russia's trade

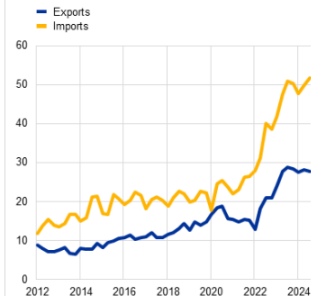
a) Share of the renminbi in settlement of China's external trade

(percentages)



b) Share of Russia's external trade with China

(percentages)



Role of geopolitics

- **Geopolitical distance** with the currency issuer reduces the use of its currency, but only recently.

Table 3: Panel regressions for export invoicing shares and geopolitical distance

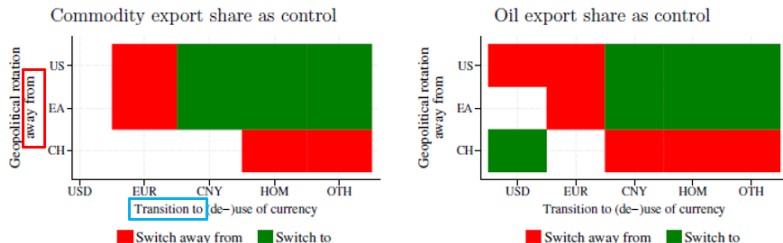
	USD		EUR		CNY	
	(1)	(2)	(3)	(4)	(5)	(6)
Lagged invoicing currency share	0.73*** (0.00)	0.75*** (0.00)	0.80*** (0.00)	0.80*** (0.00)	0.92*** (0.00)	1.02*** (0.00)
Commodity trade share	0.09*** (0.00)		-0.04** (0.04)		0.00 (0.42)	
Oil trade share		0.17*** (0.01)		-0.17*** (0.00)		-0.01* (0.06)
Geopolitical distance to invoicing currency issuer	1.06 (0.15)	-0.51 (0.54)	0.09 (0.85)	-0.28 (0.62)	-0.08** (0.02)	-0.05 (0.23)
Geopolitical distance to invoicing currency issuer × post-2021	-0.12 (0.68)	-0.36* (0.09)	-0.44* (0.06)	-0.77*** (0.00)	-0.05 (0.14)	-0.03 (0.11)
Standard controls	✓	✓	✓	✓	✓	✓
Year FEs	✓	✓	✓	✓	✓	✓
Within R-squared	0.65	0.65	0.77	0.78	0.83	0.96
Observations	1278	1307	1284	1303	696	704
Countries	113	111	111	110	89	88
p-value full effect	0.13	0.34	0.52	0.08	0.01	0.04

Boz, Emine, Anja Brügger, Camila Casas, Georgios Georgiadis, Gita Gopinath, and Arnaud Mehl (2025). "Patterns of Invoicing Currency in Global Trade in a Fragmenting World Economy", IMF working paper 25/178

Invoicing rotation

- Reduction of dollar / euro use associated with and increase in Renminbi, own, and other currencies.

Figure 12: Geopolitics and export invoicing currency



Boz, Emine, Anja Brügger, Camila Casas, Georgios Georgiadis, Gita Gopinath, and Arnaud Mehl (2025). "Patterns of Invoicing Currency in Global Trade in a Fragmenting World Economy", IMF working paper 25/178

Which currency matters for international trade

- Estimate pass-through of exchange rate into import prices, with bilateral data.
 - Exchange rate vis-à-vis the exporter's currency.
 - Exchange rate vis-à-vis the dollar.
- Stronger sensitivity to the exchange rate vis-à-vis the dollar, and to some extent vis-à-vis the euro.
- Similar for trade volumes: more affected by the dollar exchange rate.
- Invoicing affects how a currency depreciation moves the terms-of-trade (ratio of import to export prices, in exporter's currency):
 - Positive under PCP (prices set in exporter's currency, imports more expensive).
 - Negative under LCP (set in importer's currency. domestic currency value of exports higher).
 - Zero under DCP (all exports set in dollar).
- A global dollar appreciation reduces trade between non-US countries under DCP.

Exchange rate pass-through to import prices

- **Bilateral FX only**: high import prices when the importer's currency depreciates relative to the exporters.
- Add the **dollar exchange rate**: stronger role than the bilateral one.
- Interact with **share of imports invoiced in dollar**: larger role for the dollar exchange rate.

Table 1
Exchange rate pass-through into import prices.

	(1) Δp_{ijt}	(2) Δp_{ijt}	(3) Δp_{ijt}	(4) Δp_{ijt}	(5) Δp_{ijt}	(6) Δp_{ijt}
Δe_{ijt}	0.733*** (0.00)	0.177*** (0.00)	0.248*** (0.00)	0.734*** (0.00)	0.318*** (0.00)	0.436*** (0.00)
$\Delta e_{ijt} \times S_{ijt}^d$			-0.126*** (0.00)			-0.268*** (0.00)
$\Delta e_{\$jt}$		0.765*** (0.00)	0.609*** (0.00)		0.600*** (0.00)	0.231*** (0.00)
$\Delta e_{\$jt} \times S_{ijt}^d$			0.271*** (0.00)			0.656*** (0.00)
Observations	54,806	54,806	48,633	54,806	54,806	48,633
R-squared	0.239	0.271	0.284	0.258	0.287	0.400
Dyads	2,791	2,791	2,451	2,791	2,791	2,451

Notes: The first (last) three columns present unweighted (trade-weighted) regressions. All regressions include two ΔER and ΔPPI lags, and time and dyad fixed effects. Standard errors clustered by dyad. p-values in parentheses. *** p<0.01, ** p<0.05, *p<0.1.

Boz, Emine; Camila Casas, Georgios Georgiadis, Gita Gopinath, Helena Le Mezo, Arnaud Mehl, and Tra Nguyen (2022). "Patterns in Invoicing Currency in Global Trade", *Journal of International Economics* 136.

Pass-through: bilateral, dollar, euro

- Euro effect less robust than the dollar effect.

Table 2
Dollar vs. Euro Exchange Rate Pass-Through into Prices.

	(1) $\Delta p_{ij,t}$	(2) $\Delta p_{ij,t}$	(3) $\Delta p_{ij,t}$	(4) $\Delta p_{ij,t}$	(5) $\Delta p_{ij,t}$	(6) $\Delta p_{ij,t}$
$\Delta e_{ij,t}$	0.287*** (0.00)	0.179*** (0.00)	0.467*** (0.00)	0.445*** (0.00)	0.316*** (0.00)	0.698*** (0.00)
$\Delta e_{ij,t} \times (S_j^{\$} + S_j^{\epsilon})$			-0.317** (0.03)			-0.514** (0.03)
$\Delta e_{sj,t}$		0.523*** (0.00)	0.233*** (0.00)		0.590*** (0.00)	0.223*** (0.00)
$\Delta e_{sj,t} \times S_j^{\$}$			0.482*** (0.00)			0.743*** (0.00)
$\Delta e_{ej,t}$	0.488*** (0.00)	0.156*** (0.00)	0.039 (0.44)	0.162* (0.07)	-0.200 (0.11)	-0.211** (0.03)
$\Delta e_{ej,t} \times S_j^{\epsilon}$			0.746*** (0.00)			0.694*** (0.00)
Observations	40,904	40,904	36,090	40,904	40,904	36,090
R-squared	0.081	0.088	0.104	0.072	0.083	0.127
Dyads	2,739	2,739	2,399	2,739	2,739	2,399

Notes: The first (last) three columns present unweighted (trade-weighted) regressions. All regressions include two ΔER and ΔPPI lags, dyad fixed effects and global controls. Standard errors clustered by dyad. p-values in parentheses. *** p<0.01, ** p<0.05, *p<0.1.

Boz, Emine; Camila Casas, Georgios Georgiadis, Gita Gopinath, Helena Le Mezo, Arnaud Mehl, and Tra Nguyen (2022). "Patterns in Invoicing Currency in Global Trade", *Journal of International Economics* 136.

- Size of DCP is sensitive to the specification of the model (Gagnon and Sarsenbayev 2023).
- Regress change in export deflator on changes in local PPI; destinations PPI, and dominant (dollar or euro) PPI (all prices expressed in exporter's currency).
- Bigger role for PCP and LCP than for DCP.
- Difference from Boz et al, 2020 reflects:
 - Weighting by size of trade: DCP is mostly for small values.
 - Impact effect vs. effect after a year: DCP weaker as time goes by.
 - Inclusion of dominant PPI to control for inflationary environment.
 - Inclusion of Euro as dominant for Europe and Northern Africa.

Sensitivity to specification

- Smaller share of DCP with longer horizon, trade-weighting, and control for PPI.

Table 2
Comparison with [Gopinath et al. \(2020\)](#).

Variable	This paper		Gopinath et al. (2020) dataset				
	Baseline (1)	US PPI (2)	Unweighted regression	Trade-weighted regression			
			As reported (3)	As reported (4)	Sum of coefficients (5)	Unrestricted sum of coefficients (6)	Unrestricted sum without second lag (7)
ΔPCP	0.52** (0.05)	0.54** (0.05)	0.14** (0.02)	0.31** (0.04)	0.45** (0.06)	0.40** (0.06)	0.42** (0.06)
ΔLCP	0.32** (0.09)	0.48** (0.10)	0.05** (0.01)	0.07** (0.03)	0.08** (0.02)	0.41** (0.06)	0.59** (0.19)
ΔDCP	0.16* (0.08)						
$\Delta DCPUS$		-0.02 (0.10)	0.78** (0.01)	0.58** (0.04)	0.44** (0.07)	0.16 (0.10)	-0.03 (0.23)
Observations	891	891	46,820	46,820	46,820	42,303	44,131
Within R ²	0.906	0.903	0.398	0.371	0.371	0.381	0.420

* $p < 0.05$, ** $p < 0.01$.

Note: Estimates in columns 1, 2, and 7 are the sums of coefficients on the current value and one lag of the independent variables, while estimates in columns 5 and 6 additionally incorporate the second lag of the independent variables. Columns 3 and 4 report only contemporaneous coefficients. Standard errors (in parentheses) are clustered by country. For coefficient sums, standard errors are computed from the estimated coefficient covariance matrix. Within R²s refer to the percent of variation explained after removing country effects.

Sources: Authors' calculations using data described in appendix A and data provided by [Gopinath et al. \(2020\)](#).

Gagnon, Joseph, and Madi Sarsenbayev (2023). "Dollar not so dominant: Dollar invoicing has only a small effect on trade prices", *Journal of International Money and Finance* 137.

INVOICING OF SERVICES

Beyond trade in goods

- Services are a sizable share of trade (one quarter), and growing.
- Limited evidence of invoicing. Novel firm-level data for Portugal, 2014-2019 (Amador et al. 2024).
- Euro is a large share of invoicing, but firm-specific factors play a major role. Less euro use for large firms, and firms with more imported inputs invoiced in FX.
 - Import intensity: $\text{FX invoiced inputs} / \text{total cost}$.
- Within foreign currencies, (vehicle) dollar use is lower for large firms, and larger firms with dollar inputs.
- Compared with goods exports, less use of dollar in services.
 - This reflects higher labor share, and lower openness in services.

Use of non-Euro (dummy)

- More use for **larger firms**, and for firms with more **imported inputs**.

	(1)	(2)	(3)	(4)
S_f (ln turnover)	0.014** (0.006)			
S_f (ln employees)		0.017** (0.007)		
S_f (ln exports)			0.016*** (0.006)	0.041*** (0.016)
I_f^X	1.304*** (0.167)	1.311*** (0.161)	1.253*** (0.156)	
I_f^E				-0.034 (0.042)
Product-country FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
No. of observations	314491	314491	314491	314491
Adjusted R2	0.345	0.346	0.346	0.280

Standard errors in parentheses
 * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 2. Currency choices in extra-EU exports (EUR vs foreign currencies): firm-level determinants

Amador, João, Joana Garcia, Arnaud Mehl, and Martin Schmitz (2024). "Dominant currency pricing in international trade of services", CEPR discussion paper 19020

Dollar use: goods vs. services

- Less **use of dollar in services**, unless **wages** are small part of cost, or the sector is **more open**. Goods and services are then similar.

	(1)	(2)	(3)	(4)
Service _p (S)	-0.077*** (0.022)	-0.103*** (0.021)	-0.094*** (0.022)	-0.115*** (0.021)
Service _p * Low Wage Share _p (S*W)		0.083*** (0.013)		0.079*** (0.012)
Service _p * High Openness _{pc} (S*O)			0.060*** (0.017)	0.047*** (0.016)
Country-year FE	Yes	Yes	Yes	Yes
No. of observations	10541	10541	10541	10541
Adjusted R2	0.120	0.145	0.126	0.148
Test S+S*W=0 (p-value)		0.4273		
Test S+S*O=0 (p-value)			0.2077	
Test S+S*W+S*O=0 (p-value)				0.6931

Standard errors in parentheses
 * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5. Share of exports in USD: services vs. goods

Amador, João, Joana Garcia, Arnaud Mehl, and Martin Schmitz (2024). "Dominant currency pricing in international trade of services", CEPR discussion paper 19020

ENDOGENOUS INVOICING

Endogenous invoicing choice

- Previous models assume DCP as a parameter. Can it emerge endogenously (Mukhin 2022)?
- General equilibrium model, with following dimensions:
 - Openness γ (domestic bias in consumption $1 - \gamma$).
 - Inputs include labor (share $1 - \phi$) and other goods (intermediates, share ϕ).
 - Varying markup (Kimball aggregator, not CES). Optimal prices has weight $1 - \alpha$ on marginal cost, α on competitors' price (CES $\alpha = 0$).
- Exporter j selling in country i would like to set a flexible price $\tilde{p}_{ji}$ in destination currency i .
- With preset price in currency k , the price corresponds to the expected price in currency k :

$$\bar{p}_{ji}^k = E(\tilde{p}_{ji} + e_{ki})$$

- Best invoicing currency k minimizes the variance of the flexible price expressed in that currency, $\tilde{p}_{ji} + e_{ki}$ (i.e. flexible price most stable).
 - Assume invoicing in one currency, no basket.

Factors driving prices

- The optimal flexible price reflects the marginal cost, and the price of competitors selling in country i :

$$\tilde{p}_{ji} = (1 - \alpha)(mc_j + e_{ij}) + \alpha p_i$$

- The marginal cost reflects wages and the price of goods used as inputs (included imported goods):

$$mc_j = (1 - \phi) w_j + \phi p_j$$

- The consumer price basket reflects domestic goods (with home bias $1 - \gamma$) and imported goods:

$$p_j = (1 - \gamma) p_{jj} + \gamma \int_0^1 p_{ij} di$$

- These drive invoicing:
 - PCP if limited complementarities α , limited intermediate inputs ϕ , or limited openness γ .
 - LCP if more complementarity α , or limited openness γ .

Invoicing choice

- Impact of various structural parameters.

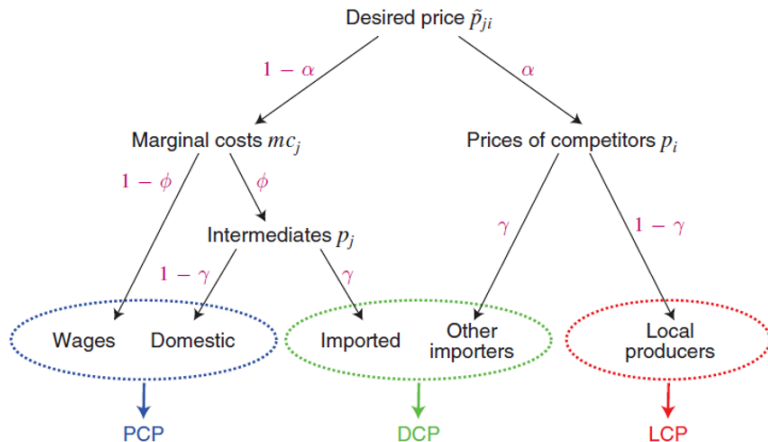


FIGURE 1. OPTIMAL PRICE AND THE CURRENCY CHOICE OF AN INDIVIDUAL FIRM

Mukhin, Dmitry (2022) "An Equilibrium Model of the International Price System", *American Economic Review*.

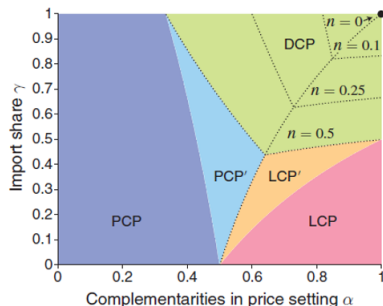
Why the dollar?

- A dominant currency can emerge, but which one?
- Large economy is favored: exporters selling in that country face a larger domestic competition.
- Stable economy is favored: lower volatility of monetary shocks (on asset returns) leads to more stable exchange rate.
- Calibrated model shows a good fit for the role of the dominant currency.
 - Caveat: does not look at dollar vs. euro.

Role of size and stability

- DCP only with strategic complementarities ($\alpha > 0$), if one country is larger (high n), or if one country is more stable (low ρ).

Panel A. Large economy advantage



Panel B. Anchor currency advantage

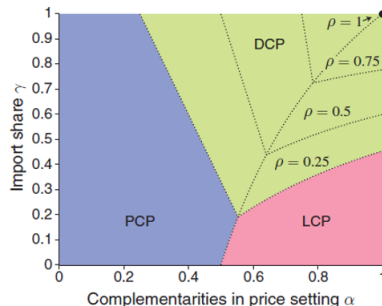


FIGURE 2. CURRENCY CHOICE: FLEXIBLE PRICE LIMIT

Notes: The figure shows equilibria for $\lambda \rightarrow 0$, $\phi = 0.5$ and (i) $\rho = 1$, different values of n ; (ii) $n = 0$, different values of ρ . PCP' (LCP') denotes the region where small countries set prices in producer (local) currency when trading with each other and use dollars when trading with the United States.

Mukhin, Dmitry (2022) "An Equilibrium Model of the International Price System", *American Economic Review*.

DEBT COST CHANNEL

Impact of foreign currency debt

- Dominant role of the dollar in funding.
- A depreciation of a country's currency against the dollar raises the (domestic currency) cost of dollar denominated debt (Casas, Meleshchuk, and Timmer 2022).
- Challenge: isolate the role of funding cost from other changes occurring at a time of recession.
- Case study of large Colombian peso depreciation in 2015.
 - Not domestically driven, instead reflect oil price (Colombia is an exporter).
 - Credit registry data on individual loans taken by firms.
 - Measure of leverage in foreign currency.
 - Measure of liquidity shocks (dollar loans coming due when the peso depreciates), and wealth shock (increase in peso value of dollar loans with long maturity).

Who suffers from higher funding costs?

- The peso depreciation leads to a reduction of imported inputs by Colombian firms.
 - Reflects liquidity and wealth shocks.
- Effect seen only for firms that do not export.
 - Exporters have earnings in dollar (used as invoicing currency of exports), and are hedged.
- Effect muted for firms that hedge exchange rate exposure through dollar assets (largest effect) or derivatives.

Effect on input imports

- **Decrease of imports** for firms with high FX leverage, or exposure to liquidity or wealth shocks.
- Concentrated among **non-exporters**, no effect for **exporters**.

Table 4: Imports

	ln (imports)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Post × FCL	-0.106*** (0.025)	-0.215*** (0.047)	-0.018 (0.023)						
Post × LS				-0.046** (0.020)	-0.069* (0.034)	-0.024 (0.021)			
Post × WS							-0.059*** (0.020)	-0.120*** (0.032)	-0.009 (0.024)
Sample	All	Non-X	X	All	Non-X	X	All	Non-X	X
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
N	524,943	350,934	174,009	524,943	350,934	174,009	524,943	350,934	174,009

Casas, Camila, Sergii Meleshchuk, and Yannick Timmer (2022) "The Dominant Currency Financing Channel of External Adjustment", International Finance discussion paper 1343.

Effect of hedge

- Firms with a **hedge** (dollar assets, or derivatives) are less sensitive to their financial exposure.

Table 7: Hedging and Imports

	ln(imports)								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Post × FCL	-0.150*** (0.045)	-0.270*** (0.062)	0.002 (0.046)						
Post × FCL × Hedge	0.140** (0.051)	0.246*** (0.079)	0.000 (0.054)						
Post × LS				-0.071** (0.030)	-0.085** (0.039)	-0.049 (0.037)			
Post × LS × Hedge				0.086** (0.040)	0.105 (0.071)	0.059 (0.046)			
Post × WS							-0.084*** (0.029)	-0.131*** (0.036)	-0.020 (0.042)
Post × WS × Hedge							0.107*** (0.038)	0.140* (0.071)	0.044 (0.049)
Sample	All	Non-X	X	All	Non-X	X	All	Non-X	X
Firm FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
Time FE	Y	Y	Y	Y	Y	Y	Y	Y	Y
N	524,943	350,934	174,009	524,943	350,934	174,009	524,943	350,934	174,009

Casas, Camila, ~~Sergii Meleshchuk~~, and Yannick Timmer (2022) "The Dominant Currency Financing Channel of External Adjustment", International Finance discussion paper 1343.

CURRENCY IN BONDS

Composition of international bonds

- Home currency bias in bonds investing: investors purchase foreign bonds, but denominated in their own currency. The dollar is the exception (Maggiore et al. 2020).
- Detailed database of outstanding stocks and issuance of international bonds (Pradhan et al 2025).
 - Issued if any of: a) under a foreign law, b) listed on a foreign exchange, c) registered in a foreign country.
- Dollar share goes into waves: decrease in early 2000's ("euro moment"), then increase.
 - Long views show a decrease in 1970's, and two waves of (temporary) increases: early 1980's, and late 1990's.
 - In part driven by exchange rate valuation effect, or by composition effect (more issuance in dollar-issuing countries), but only partially.
- Rising role of financial centers. Moving from residence to nationality data shows a larger role of emerging economies.

● Movements of the dollar share, without trend.

Figure 1. Outstanding stocks of IDS by currency of denomination, 2000-2024 (shares shown in percent)

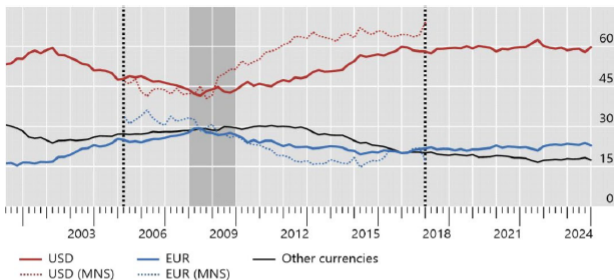
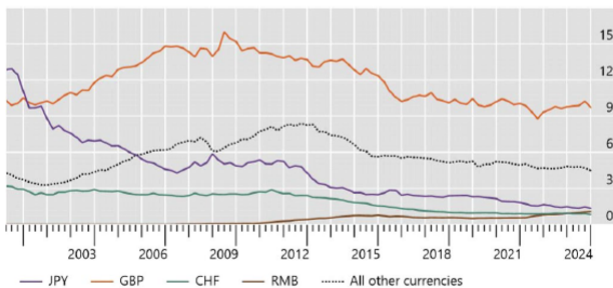


Figure 2. Outstanding stocks of IDS by currency of denomination (other than the dollar and the euro), 2000-2024 (shares shown in percent)



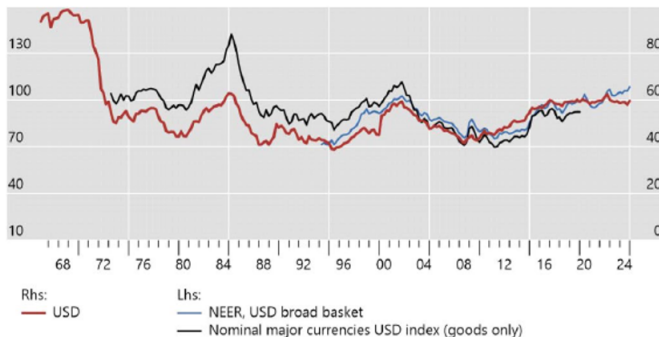
Pradhan, Swapan-Kumar, Eswar Prasad, Elod Takats, and Judit Temesvary (2025). "Dollarization Waves: New Evidence from a Comprehensive International Bond Database", Board of Governors of the Federal Reserve System International Finance Discussion Paper 142

Long perspective

- Waves of increases in the 1980's and 1990's, in part due to a strong dollar valuation effect.

Figure 3. A historical perspective on outstanding stocks of dollar-denominated IDS, 1966-2024

In percent of total in all currencies

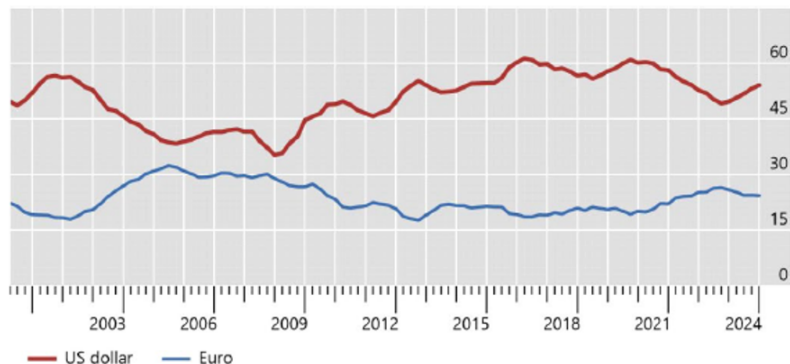


Pradhan, Swapan-Kumar, Eswar Prasad, Elod Takats, and Judit Temesvary (2025). "Dollarization Waves: New Evidence from a Comprehensive International Bond Database", Board of Governors of the Federal Reserve System International Finance Discussion Paper 142

- Pattern seen in outstanding stocks is also observed in issuance of new bonds.

Figure 4. Gross issuance of international debt securities by currency

Four-quarter moving average, in USD billion

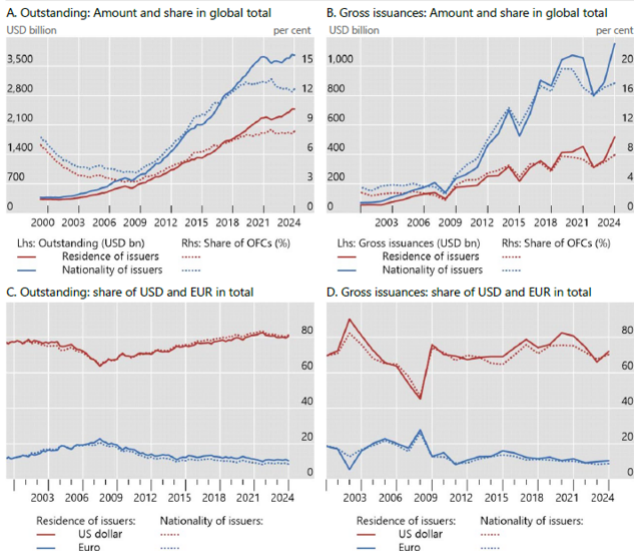


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- Much larger figures for emerging economies on a **nationality** than **residence** basis (i.e. issuance through financial centers).

Figure 10. International debt securities—issuers in major emerging market economies

By residence and by nationality basis



Pradhan, Swapan-Kumar, Eswar Prasad, Elod Takats, and Judit Temesvary (2025). "Dollarization Waves: New Evidence from a Comprehensive International Bond Database", Board of Governors of the Federal Reserve System International Finance Discussion Paper 142

PROSPECTS FOR DOLLAR DOMINANCE

Can the dollar lose its primacy?

- Assess the dollar role in a dynamic general equilibrium model, with the presence of yuan (Chahrour et al 2024).
- Joint determination of currency use in international portfolio and trade invoicing (reinforce each other, possibility of multiple steady states).
- Robust dollar position.
 - Tariffs do not eliminate it
 - Yuan promotion (removal of capital controls), or US sanctions could, but only if maintained over a decade or more.